

Revision Lab Task

Q1 : You are developing a simple Library Management System in C.

The project should contain the following files:

main.c
book.c
issue.c
library.h

Requirements:

In **book.c**

Write a function to add a book (store name and ID).

In **issue.c**

Write a function to issue a book to a student.

In **main.c**

Create a simple menu:

- Add Book
- Issue Book
- Exit

Task:

Write complete C logic in all files.

Create a Makefile that:

Compiles all .c files

Creates an executable named `library`

. Has a `clean` target

Q2. You have been assigned to prepare a structured workspace for a project audit .Create a main directory named `Project_Audit` and inside it create two subdirectories called `Reports` and `Logs`. Navigate properly between directories and verify your current location where required. Inside the `Reports` directory, create three files named `report1.txt`, `report2.txt`, and `summary.txt`. Insert today's date into each report file and add at least five lines of sample content in `report1.txt` and three lines in `report2.txt`. From `report1.txt`, extract only those lines containing the word "error" and store them inside `summary.txt`. Display the total number of lines, words, and characters in both report files, and list all files in long listing format. Modify the permissions of `summary.txt` so that the owner has full access while others have read-only access. Finally, display your current working directory, current date and time, overall disk usage in human-readable format, disk usage for the current user (if applicable), and list the files in the current directory.

Use only basic Linux commands such as `mkdir`, `touch`, `echo`, `grep`, `ls`, `ls -l`, `wc`, `date`, `pwd`, `chmod`, `df -h`, `df -u`, and `cd`.

Q3. You are required to write a shell script named `result.sh` that evaluates a student's academic performance. The script should ask the user to enter the student's name and marks obtained in three subjects. It should then calculate the total marks and average. Based on the average, assign a grade according to the following criteria:

A if average is 80 or above

B if average is 70 or above

C if average is 60 or above

D if average is 50 or above

F if below 50

Finally, the script should display the student's name, total marks, average, and grade in a properly formatted output. Use basic shell scripting concepts such as variables, user input, arithmetic operations, and conditional statements only. The script must run successfully from the terminal.

Q4: You are required to write a shell script named `analyze.sh` that demonstrates the use of environment variables and command-line arguments. The script should display the script name, the total number of arguments passed, and all arguments provided by the user. It should also display the first and second arguments separately (if provided). In addition, the script must print important environment-related information such as the current user, home directory, and present working directory using appropriate variables. If no arguments are passed, the script should display a meaningful message. Use special variables such as `$0`, `$1`, `$2`, `$#`, and `$@` along with environment variables within the script. The script must execute correctly from the terminal when arguments are passed.